ChatGPT

General-Purpose Deep Research Workflow and Reproducible Evidence-Synthesis Template

Executive summary

The **research topic, specific research question, jurisdiction, population, time horizon, language restrictions, budget, intended audience, and decision context are unspecified**. Accordingly, the framework below is designed as a reusable protocol that can be instantiated for academic, industry, technology, legal/regulatory, or public-policy research.

The core principle is to separate five activities that are often blurred together: **question definition, evidence discovery, evidence appraisal, synthesis, and decision-oriented communication**. For rigorous work, the question and eligibility rules should be specified before intensive searching; searches should be documented well enough to rerun; primary and official sources should be preferred over summaries; extraction and quality appraisal should be structured; and conclusions should distinguish what the evidence establishes from interpretation, extrapolation, and remaining uncertainty. This broadly follows the logic of the current Cochrane Handbook, PRISMA 2020/PRISMA-S, and GRADE, while adapting it beyond clinical research. [1]

A useful end-to-end workflow is:

Scope → protocol → source map → reproducible searches → screening → structured extraction → quality/bias appraisal → quantitative or narrative synthesis → contradiction analysis → uncertainty assessment → deliverables → archived evidence package.

Where a formal systematic review is appropriate, PRISMA 2020 provides a 27-item reporting framework and official flow-diagram templates; PRISMA-S adds 16 items specifically for reporting literature searches. [2] For quantitative evidence synthesis, the Cochrane Handbook covers effect measures, meta-analysis, missing evidence, bias, and interpretation; where pooling is inappropriate, SWiM provides a structured framework for reporting synthesis without meta-analysis. [3]

The attached research brief is a good example of how a generic workflow becomes domain-specific: it identifies a genre, establishes ground truth, defines exclusions and hard constraints, names open research questions, and distinguishes claims that require empirical validation from speculative claims. Those elements belong in the **research protocol** before evidence gathering begins, rather than being improvised during drafting. filecite turn0file0



Research design and methodology

A robust project starts with an **answerable question rather than a topic label**. Cochrane explicitly treats scope/question definition, inclusion criteria, searching, data collection, bias assessment, synthesis, and interpretation as separate stages; GRADE similarly emphasizes constructing an answerable question before evaluating evidence certainty. [4]

Adaptable step-by-step method

Stage	Action	Required output	Failure test
Scope	Translate the broad topic into one or more answerable questions	Question matrix	Could two researchers interpret the question differently?
Definitions	Operationally define key terms, outcomes, units, populations, technologies, jurisdictions	Definition sheet	Are ambiguous terms still doing analytical work?
Protocol	Fix eligibility criteria, source hierarchy, search approach, outcomes and synthesis method before reviewing results	Versioned protocol	Could criteria be changed after seeing favorable evidence?
Source mapping	Identify authoritative institutions, datasets, registries, original research venues and opposing perspectives	Source map	Does discovery depend mainly on generic web search?
Search	Run broad discovery searches followed by structured database searches and citation chaining	Search log	Could another researcher reproduce the query?
Screening	Apply inclusion/exclusion criteria consistently	Screening ledger + exclusion reasons	Are inconvenient studies disappearing without explanation?
Extraction	Record methods, population, outcomes, quantitative results, uncertainty, funding and limitations	Evidence table	Are conclusions being extracted without their methods/context?
Appraisal	Assess internal validity, applicability and potential conflicts/biases	Quality matrix	Are all published studies implicitly treated as equally credible?
Synthesis	Pool comparable estimates or conduct structured narrative synthesis	Analysis + synthesis matrix	Is simple vote-counting replacing effect-size reasoning?

Stage	Action	Required output	Failure test
Conflict analysis	Explain why credible sources disagree	Contradiction table	Is disagreement being hidden by averaging?
Certainty	State confidence separately for each major conclusion	Certainty statements	Is confidence higher than the underlying evidence warrants?
Reporting	Separate evidence, inference, assumptions and recommendations	Report + slides + appendix	Can the reader tell which conclusions are directly supported?
Reproducibility	Freeze searches, source list, extraction data, scripts and versions	Archival package	Could the analysis be reconstructed six months later?

For **academic research**, frame questions around population/system, intervention or exposure, comparator where applicable, outcomes, study designs, and time. PICO is well established for intervention questions, although the general principle—explicitly defining the objects being compared and the outcomes of interest—is more broadly transferable. [5]

For **industry research**, substitute a commercial structure: technology/product → customer/use case → geography → competitors/substitutes → adoption metrics → cost/economics → regulatory constraints → time horizon. Company claims should be checked against primary material such as regulatory filings rather than relying only on investor presentations or press coverage; SEC EDGAR, for example, exposes annual 10-K and quarterly 10-Q filings directly. [6]

For **policy research**, define jurisdiction, affected population, intervention/rule, counterfactual, implementation mechanism, desired outcomes, unintended effects, distributional consequences, fiscal implications, and evaluation period. Official statistical systems should generally precede third-party aggregators: the World Bank provides open development datasets, while the OECD Data Explorer is the OECD's warehouse for its statistical data and supports downloadable/programmatic access. [7]

Protocol specification

Before searching intensively, record:

Dimension	Protocol field
Research question	Exact wording
Purpose	Descriptive / causal / forecasting / evaluation / philosophical / technical / decision support
Population or system	Included entities
Geography	Included jurisdictions

Dimension	Protocol field
Time	Evidence period and publication cutoff
Outcomes	Primary and secondary outcomes
Evidence designs	Eligible research designs/source types
Primary-source rule	What counts as authoritative evidence
Exclusions	Explicit reasons for exclusion
Synthesis	Meta-analysis / descriptive statistics / narrative / mixed
Bias tools	Tools appropriate to study design
Sensitivity analyses	Pre-specified robustness checks
Stakeholders	Intended readers/decision-makers
Conflicts	Researcher/funder interests
Update policy	One-off or living review
Version	Protocol date/version/hash

For higher-rigor projects, freezing such a plan in a registry reduces ambiguity about which analytic choices preceded examination of the evidence. OSF describes preregistration as posting a time-stamped, read-only research plan before data collection or analysis, with hypotheses, exclusion rules, analyses and contingencies stated explicitly. [8]

Source hierarchy and reproducible search strategy

Evidence quality is not equivalent to publication prestige. The relevant hierarchy depends on the claim, but **original observations and authoritative records should normally anchor factual claims**.

Prioritized source hierarchy

Priority	Source type	Examples	Best use
Highest	Original official records	Statutes, regulations, court decisions, government statistics, regulatory filings, standards, official budgets	What officially happened, exists, or is required
Highest	Primary research and datasets	Original papers, trial reports, registered protocols, survey microdata, experiments, observational datasets	Empirical effects and mechanisms

Priority	Source type	Examples	Best use
Highest	Technical primary material	Specifications, source code, benchmark datasets, patents, standards documents, engineering reports	Technical capabilities and constraints
High	Systematic evidence syntheses	Systematic reviews, meta-analyses, evidence reports	Mapping mature literatures
High	Authoritative institutional analyses	National academies, statistical agencies, OECD, World Bank, major regulators	Cross-source context and policy evidence
Medium	High-quality secondary scholarship	Review articles, scholarly books, methodological papers	Theory, historical interpretation, synthesis
Contextual	Reputable journalism	Investigative or specialist reporting	Leads, chronology, interviews, developments absent from literature
Discovery only	Company marketing, advocacy material, blogs, Wikipedia, social media, LLM output	Various	Leads and terminology, not unverified final evidence

The hierarchy is **claim-specific**. A randomized trial may be superior evidence for an intervention effect but irrelevant to the legal text of a regulation; the regulation itself is the primary source for what the law says. Likewise, audited/regulatory filings are generally preferable to media summaries when documenting a public company's reported financial information. SEC EDGAR directly identifies annual and quarterly reports as 10-K and 10-Q filings. [6]

For international policy or economic comparisons, official datasets such as World Bank Open Data and OECD Data Explorer provide documented source systems rather than an extra layer of secondary re-publication. [7]

Database strategy

Use **multiple, partly overlapping discovery systems**, because no single index captures an entire literature. Cochrane explicitly warns that missing evidence can bias synthesis and notes that comprehensive searches may require bibliographic databases, registries, regulators, manufacturers, authors and sponsors. [9]

A practical map is:

Domain	Search first	Then supplement with
Biomedicine	PubMed/MEDLINE	Trial registries, regulators, Cochrane, citation chaining
General science	Discipline indexes + Crossref/ Google Scholar-style discovery	Publisher sites, repositories, references/ citations
Computing/ engineering	IEEE/ACM and original conference/ journal repositories	Standards bodies, GitHub/code, benchmark repositories
Economics	Journal databases, working-paper repositories	Statistical agencies, central banks, World Bank/OECD
Industry	Regulator filings, corporate reports	Patents, market datasets, technical papers, trade literature
Public policy	Government/legislative sites	OECD, World Bank, evaluation literature, think tanks
Legal/regulatory	Official legal databases/regulators	Law reviews and specialist commentary
Emerging topics	Preprints + original technical material	Conference proceedings, repositories, subsequent peer-reviewed versions

PubMed's current search system supports phrases, field tags, Boolean `AND/OR/NOT`, wildcards, proximity searching, publication-date restrictions, filters, search histories and MeSH-controlled vocabulary. [10]

Reproducible query construction

Build queries as **concept blocks**:

```
(CONCEPT_A_synonym1 OR CONCEPT_A_synonym2 OR "exact phrase")
AND
(CONCEPT_B_synonym1 OR CONCEPT_B_synonym2)
AND
(outcome1 OR outcome2 OR outcome3)
```

Example generic technology/policy search:

```
("generative artificial intelligence" OR "large language model*" OR LLM*)
AND
(education OR classroom OR student*)
AND
(learning OR achievement OR productivity OR assessment)
NOT
(editorial OR commentary)
```

Do not indiscriminately add `NOT`: exclusions can accidentally remove relevant records. In PubMed, Boolean operators are processed explicitly and parentheses are used to nest concepts, making the recorded final query part of the reproducibility record. [10]

For every database, save:

```
Database:
Platform/provider:
Date searched:
Coverage dates:
Exact search string:
Filters:
Number retrieved:
Export format:
Export filename:
Deduplication method:
Search author:
Notes/deviations:
```

PRISMA-S exists specifically because merely writing “we searched PubMed” is inadequate for reproducing an evidence search; its checklist covers reporting search methods in systematic reviews. [11]

Date rules

Avoid a universal “last five years” cutoff. Recommended decision rules are:

Foundational search: no arbitrary lower date limit when the history of a concept matters.

Current-state search: define a contemporary window appropriate to the rate of change, then separately search seminal earlier work.

Fast-moving technology/policy: search by both publication date and event/version date; a recent article may describe older technology or legislation.

Living questions: record a cutoff date and schedule updates rather than silently mixing evidence retrieved at different times. PRISMA has a dedicated extension for living systematic reviews and updates. [12]

Inclusion and exclusion criteria

Eligibility should be based on **relevance and evidentiary fitness**, not whether the result agrees with the hypothesis.

A reusable rule is:

> Include evidence that directly addresses a predefined question, population/system, outcome, jurisdiction and eligible design within the stated temporal scope. Exclude duplicates, clearly irrelevant records, superseded versions where a definitive version exists, sources lacking the minimum information required for appraisal, and study designs explicitly outside scope. Record a reason for every full-text exclusion.

Do not automatically exclude null results, negative findings, inconvenient outcomes, preprints, non-English material, or evidence because it is difficult to obtain; each can create systematic bias if excluded without a methodological reason. Cochrane specifically treats selective non-reporting as a potential source of bias in a review. [9]

PRISMA's official flow diagrams are a useful visual model for recording identification, screening, inclusion and exclusions. [13]

Data extraction, appraisal and evidence tables

Extraction should preserve enough context that the analyst does not have to repeatedly reinterpret the original source.

Master extraction template

Field	Extract
Record ID	Unique identifier
Citation	Author, title, year, DOI/URL/identifier
Source type	Paper, dataset, regulation, filing, report, etc.
Research question	What the source actually tests/addresses
Study/design	RCT, cohort, cross-sectional, simulation, qualitative, theoretical, etc.
Population/sample	N, geography, eligibility
Intervention/exposure	Definition, dose/intensity where relevant
Comparator	Control/reference
Outcome	Exact operational definition
Follow-up	Duration/time points
Statistical method	Model/tests/estimand
Effect/result	Estimate or main result
Uncertainty	CI/SE/credible interval/p-value as applicable
Missing data	Extent/treatment
Key finding	Concise neutral summary

Field	Extract
Limitations	Author-reported + reviewer-identified
Risk of bias	Tool/domain judgments
Funding	Funding source
Conflicts	Declared interests
Data availability	Repository/access
Code availability	Repository/version
Relevance	Direct/indirect
Reviewer notes	Interpretation issues
Verification status	One- or two-reviewer checked

Illustrative filled extraction table

The following entries are deliberately fictional; they demonstrate format rather than constitute evidence about a real topic.

Study	Year	Methods	Key findings	Limitations
Example Study A	2023	Randomized trial; 420 participants; intervention vs. standard practice; 12-week follow-up	Intervention improved the prespecified primary outcome by a modest amount; confidence interval excluded the null	Single region; short follow-up; attrition differed slightly by arm
Example Study B	2024	Prospective observational cohort; N=3,100; multivariable adjustment	Exposure correlated with better outcome after adjustment, but effect was smaller than Study A	Residual confounding; self-selected exposure; measurement error
Example Study C	2025	Multi-site qualitative study; 68 interviews; thematic analysis	Identified implementation barriers and heterogeneous stakeholder responses	Not designed to estimate causal effect; purposive sample limits generalization
Example Study D	2026	Administrative-data natural experiment; difference-in-differences	Estimated a small population-level effect with strong geographic heterogeneity	Parallel-trends assumption; treatment spillovers possible

The important point is that **“key finding” never travels without design and limitation fields**.

Bias and quality assessment

Use tools matched to design rather than assigning every paper an informal “high/medium/low quality” score. Cochrane currently recommends **RoB 2** for randomized trials and **ROBINS-I** for non-randomized intervention studies; a revised ROBINS-I version was released in 2025 with algorithmic support for judgments and additional treatment of issues such as immortal-time bias. [14]

For an evidence body rather than one study, GRADE evaluates certainty using domains including risk of bias, inconsistency, indirectness, imprecision and dissemination/publication bias, with certainty expressed in four levels: high, moderate, low and very low. [15]

A general-purpose appraisal matrix can therefore separate:

Dimension	Question
Internal validity	Could design or analysis systematically distort the result?
Precision	How wide is plausible uncertainty?
Directness	Does the evidence actually match the target question?
Consistency	Do credible sources converge?
Completeness	Could important missing evidence alter the conclusion?
Measurement validity	Does the instrument measure the claimed construct?
External validity	Does the result generalize to the intended setting?
Independence	Are financial/institutional incentives relevant?
Reproducibility	Are data/code/methods inspectable?
Temporal validity	Is the evidence still applicable today?

Bias assessments should affect interpretation rather than simply appearing in an appendix; Cochrane explicitly notes that biased individual-study results can produce misleading meta-analytic conclusions. [16]

Synthesis, meta-analysis and conflicting evidence

Quantitative synthesis

Conduct a meta-analysis only when the studies answer sufficiently comparable questions. A defensible workflow is:

1. Define the target estimand and eligible outcomes before inspecting pooled results.
2. Harmonize outcome direction, units and effect measures.
3. Separate materially different designs/populations rather than forcing them into one estimate.
4. Examine study-level risk of bias.
5. Calculate study estimates and sampling variances.

6. Select the statistical model based on the substantive model of variation, not merely on whether a heterogeneity test crosses a p-value threshold.
7. Estimate pooled effects with uncertainty.
8. Examine heterogeneity and, where meaningful, prediction intervals.
9. Conduct prespecified subgroup/meta-regression analyses only where enough evidence exists to support them.
10. Run sensitivity analyses excluding high-risk, indirect, influential or alternative-definition studies.
11. Assess missing-evidence/publication-bias concerns where the dataset makes such assessment meaningful.
12. Interpret clinical/policy/practical importance separately from statistical significance.

These steps reflect the Cochrane Handbook's separation of effect measures, bias, synthesis preparation, meta-analysis, missing evidence, certainty and interpretation. [17]

Do **not** average fundamentally different constructs merely because they can all be represented numerically.

Narrative synthesis

When meta-analysis is unjustified, use a structured evidence matrix rather than an impressionistic literature review. SWiM was developed specifically for transparent reporting of synthesis without meta-analysis. [18]

Organize evidence by predefined dimensions such as:

```
Mechanism
    ↓
Population / context
    ↓
Study design
    ↓
Direction of effect
    ↓
Magnitude
    ↓
Precision
    ↓
Risk of bias
    ↓
Applicability
```

Report ranges and study characteristics instead of phrases such as “most studies agree” whenever effect estimates can be reported directly.

Decision rules for conflicting evidence

When strong sources disagree, do **not** resolve the conflict by counting papers. Use the following priority rules:

Conflict	Decision rule
High-quality direct vs. weak indirect evidence	Give greater weight to direct, lower-bias evidence
Large vs. small studies	Consider precision, but not sample size independently of design validity
RCT vs. observational result	Ask whether they estimate the same causal question and population before privileging design
Old vs. new evidence	Prefer newer evidence only when the underlying technology/context has changed or the new methods/data are stronger
Peer-reviewed vs. preprint	Treat review status as one quality signal, not proof of truth
Company vs. independent estimate	Inspect definitions, denominators, incentives and original data
Official data vs. third-party aggregation	Prefer the underlying official series unless the third party performs a justified transformation
Multiple contradictory credible sources	Preserve the contradiction and downgrade confidence rather than inventing consensus

A useful contradiction ledger is:

Claim	Evidence supporting	Evidence opposing	Likely reason	Resolution
X produces Y	Studies A/C	Studies B/D	Different populations	Effect probably context-dependent
Technology achieves Z	Vendor benchmark	Independent test	Different benchmark definition	No directly comparable estimate
Policy reduced outcome Q	Agency evaluation	Academic quasi-experiment	Different counterfactual	Unresolved; report both

GRADE's approach is useful conceptually here because inconsistency, indirectness, imprecision and bias are treated as reasons to reduce certainty rather than reasons to suppress contradictory evidence. [19]

Deliverables, visualizations and project sizing

Deliverable templates

Suggested sizes below are **planning heuristics, not methodological standards**.

Deliverable	Suggested size	Contents
Executive summary	500–1,200 words	Question, headline findings, confidence, implications, critical uncertainties
Full concise report	3,000–6,000 words	Methods, evidence, synthesis, limitations, conclusions
Full deep-research report	7,000–15,000 words	Full methods, literature, analyses, competing interpretations
Major technical report	15,000–30,000+ words	Multiple questions, modeling, appendices, detailed provenance
Briefing deck	8–12 slides	Decision-oriented findings
Full presentation	15–25 slides	Methods, evidence and implications
Technical deck	25–40 slides	Detailed analyses and appendices
Data appendix	As required	Search logs, exclusions, extraction tables, quality ratings, calculations

A strong **executive summary** can use this template:

> **Question:** What was investigated?
> **Bottom line:** What does the strongest available evidence indicate?
> **Confidence:** High / moderate / low / very low, with reasons.
> **Key evidence:** Which primary sources most strongly determine the conclusion?
> **Disagreement:** What credible evidence points elsewhere?
> **Implications:** What follows—and what does not follow—from the findings?
> **Unknowns:** Which unresolved issues could materially change the answer?

The **full report** should normally contain: scope and definitions; methods; source-selection logic; factual baseline; evidence by research question; quantitative/narrative synthesis; contradictions; limitations; uncertainty; implications; and reproducibility appendix.

The **slide deck** should not simply compress prose. A useful sequence is:

```
Question
↓
Why it matters
↓
What evidence was searched
↓
What the strongest evidence says
↓
What disagrees
↓
What explains the disagreement
```

```
↓
Confidence / uncertainty
↓
Implications
↓
Open questions
```

Recommended visualizations

Use the visualization dictated by the analytical question:

Question	Visualization
How were sources screened?	PRISMA-style flow diagram
How do study estimates compare?	Forest plot
How does evidence quality vary?	Risk-of-bias traffic-light plot
What changed over time?	Time series with event annotations
How do scenarios compare?	Dot/interval plot rather than decorative bars
How uncertain is a forecast?	Fan chart / prediction intervals
Why do studies disagree?	Evidence matrix or subgroup forest plot
Which concepts depend on which?	Causal/concept map
How does a policy work?	Logic model
How does evidence move to a conclusion?	Evidence-to-claim graph
What areas lack research?	Evidence-gap matrix
How does an industry fit together?	Value-chain or ecosystem map

PRISMA provides official flow-diagram examples suitable as the model for source-screening graphics. [13] Cochrane's current RevMan guidance illustrates risk-of-bias “traffic light” visualization alongside forest plots. [20]

Project timelines and resources

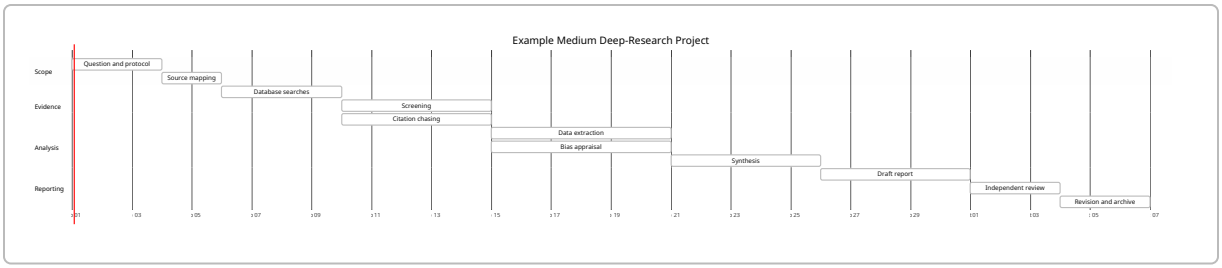
Again, these are **practical budgeting assumptions**, not universal standards.

Scale	Typical scope	Team	Calendar	Effort	Expected evidence set
Small	One tightly bounded question	1 researcher + reviewer	3–7 working days	~3–8 person-days	~20–60 core sources

Scale	Typical scope	Team	Calendar	Effort	Expected evidence set
Medium	Several linked questions	2–3 researchers	2–5 weeks	~15–40 person-days	~50–200 screened/ included sources
Large	Multi-domain or systematic assessment	3–6+ specialists	6–16+ weeks	~60–200+ person-days	Hundreds to thousands screened

These estimates assume readily accessible sources. Translation, proprietary datasets, interviews, original experiments, legal review, complex modeling, individual-participant-data meta-analysis, or manual extraction from large archival collections can increase requirements substantially.

A medium-project schedule might look like:



For high-stakes systematic reviews, independent duplicate screening/extraction is preferable where resources permit because it provides a check on human judgment. At minimum, a second reviewer should audit consequential inclusion decisions, extraction fields and claims that materially determine the conclusion.

Citation practice, reproducibility, ethics and common failure modes

Citation and claim discipline

A research report should make it possible to answer **“Which evidence establishes this exact sentence?”**

Recommended rules are:

Cite the primary source for the primary claim. Cite a statute for statutory language, original dataset for a statistic, study for its result, standard for its specification, and regulatory filing for a company's filed disclosure.

Use secondary sources for interpretation and context, not as substitutes for inspectable primary evidence when the latter exists.

Place citations adjacent to claims, not as undifferentiated clusters at the end of pages.

Distinguish four epistemic categories explicitly:

> **Observed/reported:** directly present in evidence.
> **Calculated:** derived transparently from evidence.
> **Inferred:** analyst interpretation supported by evidence.
> **Speculative:** plausible hypothesis not established by evidence.

Never convert an absence of evidence into evidence of absence unless the search/design has sufficient power to make the non-observation informative.

Keep event dates separate from publication dates. This matters especially in fast-changing technology, industry and policy research.

Reproducibility package

The minimum archival package should contain:

```
/research-project
│
├── README.md
├── protocol/
│   ├── protocol_v1.md
│   └── deviations.md
│
├── searches/
│   ├── search_log.csv
│   ├── pubmed_search.txt
│   ├── database_02_search.txt
│   └── exports/
│
├── screening/
│   ├── records.csv
│   └── exclusions.csv
│
├── extraction/
│   ├── evidence_table.csv
│   └── codebook.md
│
├── appraisal/
│   └── risk_of_bias.csv
│
├── analysis/
│   ├── scripts/
│   ├── environment.lock
│   └── outputs/
│
```

```
├── figures/
│
├── report/
│
└── CITATION.cff
```

Record software and package versions, random seeds where applicable, raw-to-processed-data transformations, database search dates, deduplication procedures, and every manual correction that materially affects results.

For long-term dissemination, a persistent research repository is preferable to an unversioned personal folder. Zenodo, operated by CERN/OpenAIRE, accepts research data, software and other scholarly artifacts, assigns DOIs to published records, supports versioning, and integrates with GitHub releases; its GitHub integration can automatically ingest enabled repository releases. [21] DOI-based archival records improve persistent identification and interlinking of research outputs. [22]

For planned studies or reviews, an immutable or time-stamped protocol registration can complement the final archive; OSF Registrations are designed to preserve a transparent record of a research plan and its subsequent updates. [8]

Uncertainty reporting

Do not collapse uncertainty into a single confidence adjective. Separate at least:

Uncertainty	Example
Sampling	Estimate has a wide interval
Measurement	Outcome proxy may poorly represent target construct
Model	Different defensible specifications produce different results
Structural	Causal mechanism is uncertain
External validity	Study population differs from target population
Publication	Negative/null results may be missing
Temporal	Older evidence may no longer describe current technology
Definition	Sources operationalize a term differently
Data provenance	Origin/transformation of values is unclear
Deep/metaphysical	Question may not be empirically decidable

This distinction prevents a precise-looking statistical interval from being mistaken for total epistemic certainty. GRADE similarly treats risk of bias, inconsistency, indirectness, imprecision and publication/dissemination concerns as distinct dimensions affecting evidence certainty. [19]

Ethical considerations

Research design should consider not only whether data are publicly accessible but whether their use is ethically defensible. Sensitive personal, medical, location, demographic, or proprietary information may require access controls, aggregation or non-release even when openness would otherwise aid reproducibility. Research archives themselves distinguish open from restricted material; Zenodo, for example, supports restricted and embargoed content rather than requiring every artifact to be made immediately public. [23]

Further recurring ethical questions include conflicts of interest, informed treatment of affected communities, disproportionate impacts across populations, dual-use risks, confidentiality, intellectual property, misrepresentation of uncertainty, and the danger of turning speculative evidence into apparently operational recommendations.

Particularly important is the distinction between **measurement capability and construct validity**. A device, model, benchmark or dataset may accurately measure *something* without having been validated as measuring the more ambitious construct an analyst assigns to it. In speculative or emerging fields, this is often the central evidentiary gap rather than a minor methodological detail. The attached brief provides a concrete example of this distinction by insisting that technical measurement claims be separated from unvalidated identity- or consciousness-level interpretations. filecite turn0file0

Common pitfalls and safeguards

Pitfall	Why it fails	Safeguard
Starting with a conclusion	Encourages confirmation bias	Lock question/criteria first
Searching only Google	Incomplete and poorly reproducible	Use domain databases + primary repositories
Treating citations as votes	Ignores methods and precision	Weight by validity/directness
“Latest = best”	New evidence may be weaker	Compare methods and context
Over-filtering by date	Deletes foundational evidence	Separate seminal/current searches
Unlogged query changes	Makes search irreproducible	Version search strings
Using abstracts only	Misses methods and caveats	Extract from complete primary source
Mixing outcomes	Creates false comparability	Prespecify constructs
Pooling heterogeneous studies	Produces meaningless averages	Test conceptual comparability first
Reporting p-values alone	Hides magnitude/uncertainty	Report effects and intervals
Ignoring null/unpublished evidence	Inflates apparent effects	Search registries/grey literature

Pitfall	Why it fails	Safeguard
Using composite “quality scores” mechanically	Conceals distinct biases	Preserve domain-level judgments
Treating a model as observation	Converts assumptions into facts	Label modeled vs. observed quantities
Unsupported precision	Implies knowledge not present	Match decimal/numeric precision to evidence
LLM-generated citations	Can introduce nonexistent/ mismatched sources	Verify every source against the original
Hiding disagreement	Produces false consensus	Maintain contradiction ledger
Confusing feasibility with desirability	Technical ability ≠ policy recommendation	Separate evidence from normative judgment
Treating speculation as a roadmap	Creates false actionability	Label speculative assumptions and enabling gaps

The broader methodological lesson is simple: **the rigor of deep research is determined less by how many sources appear in the bibliography than by whether every consequential conclusion can be traced through an explicit chain from question → source → extraction → appraisal → synthesis → uncertainty → claim.** PRISMA, Cochrane, GRADE, reproducible search logging, and persistent data/code archives provide established components for constructing that chain; the domain-specific protocol determines how they should be adapted. [24]

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[1] [4] [17] Cochrane Handbook for Systematic Reviews of Interventions | Cochrane Training
https://training.cochrane.org/handbook/current?utm_source=chatgpt.com

[2] PRISMA 2020 statement — PRISMA statement
https://www.prisma-statement.org/prisma-2020-statement?utm_source=chatgpt.com

[3] Part 2: Core methods | Cochrane Training
https://training.cochrane.org/handbook/current/part-2?utm_source=chatgpt.com

[5] GRADE Book
https://book.gradepro.org/guideline/principles-for-assessing-the-certainty-of-interventions?utm_source=chatgpt.com

[6] EDGAR Entity Landing Page
https://www.sec.gov/edgar/browse/?CIK=0001784058&utm_source=chatgpt.com

[7] World Bank Open Data | Data
https://data.worldbank.org/?utm_source=chatgpt.com

[8] Welcome to Registrations & Preregistrations! - OSF Support
https://help.osf.io/article/330-welcome-to-registrations?utm_source=chatgpt.com

[9] Chapter 13: Assessing risk of bias due to missing evidence in a meta-analysis | Cochrane Training
https://training.cochrane.org/handbook/current/chapter-13?utm_source=chatgpt.com

10 Help - PubMed
https://pubmed.ncbi.nlm.nih.gov/help/?utm_source=chatgpt.com

11 Search — PRISMA statement
https://www.prisma-statement.org/prisma-search?utm_source=chatgpt.com

12 LSR — PRISMA statement
https://www.prisma-statement.org/lsr?utm_source=chatgpt.com

13 PRISMA 2020 statement — PRISMA statement
https://www.prisma-statement.org/prisma-2020?utm_source=chatgpt.com

14 Risk of bias | Cochrane
https://www.cochrane.org/learn/courses-and-resources/cochrane-methodology/risk-bias?utm_source=chatgpt.com

15 19 GRADE Book
https://book.gradepro.org/guideline/overview-of-the-grade-approach?utm_source=chatgpt.com

16 Chapter 7: Considering bias and conflicts of interest among the included studies | Cochrane Training
https://training.cochrane.org/handbook/current/chapter-07?utm_source=chatgpt.com

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